

Helix OTA — §11.4.25 Coverage Ledger (1.0.0-MVP)

Revision: 1 **Last modified:** 2026-06-08T00:00:00Z **Authority:** Constitution §11.4.25 (Full-Automation-Coverage) + §11.4.6 (No-guessing) + §11.4.27 (100% test-type coverage) **Scope:** Go control plane (server/) + owned device-side / protocol bricks (submodules/ota-*, submodules/http3) **Maintainer:** QA lead

Post-authoring update (2026-06-08, commit 5363511). This ledger was authored from the tree just before the §11.4.85 stress+chaos suite landed. The “Stress” and “Chaos” cells marked MISSING for the operational surface (groups / reads / membership / repo-fault) are now **COVERED** by server/internal/api/resilience_test.go (race-clean; evidence in docs/qa/20260608-stress-chaos/). Performance/NFR + adversarial-security suites were in flight the same session — see their qa dirs once committed.

Anti-bluff note (§11.4.6 / §11.4.25). Every COVERED cell below cites the test file (and, where useful, the function) that actually exists in the tree as of this revision. A cell is marked COVERED **only** when a test exercising that feature is present. Advanced test types not yet landed are marked MISSING or “in progress (this session)” — they are **not** claimed as coverage. On-device (RK3588 / Orange Pi 5 Max) validation is the one class blocked on hardware; it is marked BLOCKED-HW everywhere it applies and is **never** counted as automated coverage.

Legend

Mark	Meaning
COVERED	A real test exercising the feature exists; file cited.
PARTIAL	Some paths covered, notable gaps remain; file cited.
MISSING	No test of this type for this feature exists yet.
N-A	Test type does not apply to this feature.
BLOCKED-HW	Requires RK3588 hardware; cannot be automated tonight.
in progress	Being authored this session; not yet landed — do NOT count.

Test-type columns: **U**=unit, **I**=integration (httptest / real-Postgres), **E2E**=live-server end-to-end, **Sec**=security, **Str**=stress, **Ch**=chaos, **Perf**=performance, **Chal**=HelixQA Challenge.

1. Control plane (server/) — feature × test-type matrix

Feature	U
Auth (login / JWT / refresh rotation / RBAC)	COVERED api/handlers_auth_test.go:TestLogin,TestRefreshRotation,TestRBAC
Artifact upload + signature verify	COVERED api/handlers_artifact_test.go, handlers_artifact_parts_test.go
Releases (create / get / monotonicity / list)	COVERED api/handlers_release_test.go
Deployments (all-targets / conflict / unknown-release)	COVERED api/handlers_deployment_test.go
Staged rollout engine (phases / halt-on-breach)	COVERED rollout/store_test.go:TestEngineDrivesPhasesToCompletion,TestHaltOnBreach
Recall / rollback	COVERED api/handlers_recall_test.go:TestRecallRecordsRollback,TestRollback
Audit trail	COVERED api/handlers_audit_test.go:TestAuditRecordsSuccessfulMutation,TestAuditRecordsFailure,TestAuditRecordsFilter,TestAuditRecordsWiden,TestAuditRecordsSinceUntilFilter
Telemetry reads + overview	COVERED api/handlers_telemetry_test.go, handlers_widen_test.go:TestTelemetryReads,TestTelemetryOverview
Telemetry ingest (client)	COVERED api/handlers_client_test.go:TestClientTelemetryIngest,TestClientTelemetryOverview
Device register + status	COVERED api/handlers_device_test.go (register / validation / conflict / idempotent)
Device groups + members	COVERED api/handlers_group_test.go:TestGroupCRUDLifecycle,TestGroupMembers
Client update decision (offer / 200 /	COVERED api/handlers_client_test.go (offers-delta / 200-behind / 204-on-target)

Feature	U
204 / short-circuit)	
Anti-downgrade guard	COVERED api/handlers_client_antidowngrade_test.go:TestClientUpdateNeverOffersDowngrade
Branches / widen (advanced upload + filter edge cases)	COVERED api/handlers_branches_test.go, handlers_widen_test.go
Delta register / select	COVERED api/handlers_delta_test.go:TestDeltaRegisterAndFind,TestDeltaSelect
Persistence seam — in-memory Repository	COVERED store/memory_test.go, store/contract_test.go:TestMemoryRepository
Persistence seam — pgx/PostgreSQL Repository	COVERED (same contract)
Health / readiness probes	COVERED health/health_test.go, api/handlers_health_test.go
Config loading	COVERED config/config_test.go:TestLoadDefaults,TestLoadOverrides,TestLoadErrors
Transport — HTTP/3 (QUIC) + H2 fallback	COVERED transport/transport_test.go:TestDualTransportServesH3AndH2
Response compression — Brotli / gzip / identity	COVERED api/middleware_compression_test.go (brotli-negotiate / gzip-fallback / identity)
NFR / load (latency / RPS / p99)	N-A

2. Device-side / protocol bricks (submodules/) — feature × test-type matrix

Feature	U	I
Protocol types / enums / payload / validation	COVERED ota-protocol/{enums,payload,types,validate}_test.go (round-trips, invalid-enum-rejected, descriptive errors)	COVERED
Artifact validator (hash / sig / version / target / metadata)	COVERED ota-artifact-validator/validator_test.go (TestValidateHash/Signature/Version/Target/Metadata/FailFast)	COVERED
Rollout engine (cohort hashing / decide / phases / halt / window)	COVERED ota-rollout-engine/{cohort,decide,engine}_test.go (determinism, boundaries, monotonic growth, full progression, halt-idempotent, window-hold, post-boot-abort)	COVERED
Telemetry schema (codec / events / health derivation)	COVERED ota-telemetry-schema/{codec,event,health}_test.go (batch round-trip, invalid-event-reject, health counts/rates/verdict/thresholds)	COVERED health_t
HTTP/3 server brick (config / TLS / lifecycle / cross-backend)	COVERED http3/pkg/server/server_test.go (config-validate, forces-TLS1.3+H3-ALPN, start-twice, idempotent-shutdown)	COVERED (real round
Device delta-apply decision (agent)	COVERED ota-android-agent/.../delta/DeltaApplyDecisionTest.kt (12 @Test: base-version/hash match, malformed gates, ordering, mutation-immunity)	N-A (pure
Verify-before-apply (agent: hash + signature)	COVERED ota-android-agent/.../verify/VerifyBeforeApplyTest.kt (9 @Test: hash/sig accept+reject, ordering, mutation-immunity)	N-A
Agent poll state machine + jitter	COVERED ota-android-agent/.../poll/PollStateMachineTest.kt (10), JitterTest.kt (6)	N-A
Agent protocol codecs (round-trip)	COVERED ota-android-agent/.../protocol/CodecRoundTripTest.kt (11 @Test)	N-A
update_engine bridge (apply request /		N-A

Feature	U	I
status / error / payload props)	COVERED ota-update-engine-bridge/core/src/test/.../{ApplyRequest,EngineStatus,EngineError,PayloadProperties}Test.kt	
A/B update + AVB/dm-verity + auto-rollback (on-device)	BLOCKED-HW	BLOCKED

3. Biggest gaps (most → least critical)

1. **Stress + Chaos test types are essentially absent project-wide (§11.4.85).**
Across both the control plane and the bricks, the Str and Ch columns are MISSING almost everywhere. There is no sustained-load suite, no concurrent-contention suite, and no fault-injection (process-death / network-fault / resource-exhaustion / state-corruption) suite wired into any gate. The only load-adjacent asset is `server/tools/loadtest/main.go`, which **measures** but **asserts nothing** and is not part of an automated gate.
2. **Performance / NFR has no automated assertion gate.** `loadtest` reports p50/p90/p99/RPS from real round-trips (honest, anti-bluff by design) but deliberately asserts no target. No CI/pre-build step compares measured numbers against stated NFR targets, so a regression in latency or throughput would not fail any gate.
3. **Security coverage is real but narrow — it is RBAC/validation-shaped, not adversarial.** Genuine security invariants ARE covered (the artifact trust-boundary `TestArtifactUploadIgnoresRequestSuppliedPubkey`, the anti-downgrade guard, the HTTP/3 TLS1.3 floor, RBAC-forbidden paths on every mutating route). What is MISSING: a dedicated security test type — JWT tamper/expiry/replay, fuzzing of request bodies beyond `http3/.../fuzz_test.go:FuzzConfigValidate`, injection/DoS, and authz-matrix sweeps. Today these live as side-assertions inside unit/integration tests, not as a first-class Sec suite.

Secondary: **E2E breadth.** The live-server E2E (`challenge_operational.sh`) is strong for `auth/device/group/audit/telemetry/rollout-gates` but does NOT drive the full signed-artifact → release → deployment → rollout-evaluate pipeline end-to-end (it SKIPS-with-reason at step 13 per §11.4.3). Releases, recall, delta-select, and client-update decision have no live-server E2E.

4. On-device validation (the one hardware-blocked class)

All RK3588 / Orange Pi 5 Max on-device classes — real `update_engine` A/B apply, AVB/dm-verity verification, auto-rollback, and live agent poll→download→verify→apply→report against a running server — are BLOCKED-HW and are **not**

counted as automated coverage anywhere above. The agent-side **logic** for those flows (delta decision, verify-before-apply, poll state machine, bridge DTOs) IS unit-covered in submodules/ota-android-agent and submodules/ota-update-engine-bridge; only the on-hardware execution is blocked. Per §11.4.3 this is the correct SKIP/BLOCKED posture — never a PASS-by-default.